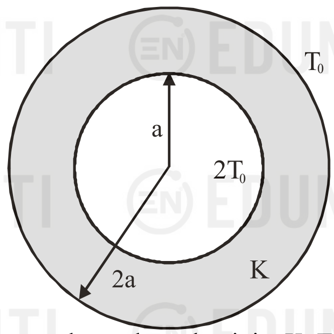
SUBJECTIVE TYPE

1. A uniform rod of cross section area 1 cm^2 and length 1m is initially at a temperature of 0^0C. It is then kept between 2 reservoirs of temperature 100^0C and 0^0C. Specific heat capacity is 10 J/kg^{0}C and linear mass density is 2kg/m. Find the

(a) temperature gradient along the rod in steady state
(b) total heat absorbed by the rod to reach steady state

2. A hollow spherical shell of inner radius a and outer radius 2a is made of a uniform



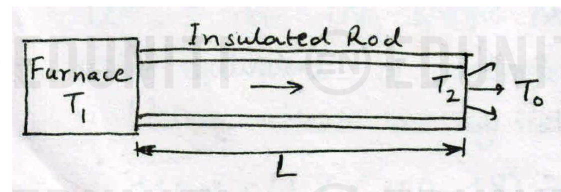
material of constant thermal conductivity K. The temperature within the shell is maintained at $2T_0$ and outside the ball it is T_0. Assuming steady state condition, find
(a) the rate at which heat flows out of the ball
(b) the temperature at r = 3a/2, where r is radial distance from the centre of shell.

3. A cylindrical block of length 0.4 m and area of cross–section 0.04 m^2 is placed coaxially on a thin metal disc of mass 0.4 kg and of the same cross–section. The upper face of the cylinder is maintained at a constant temperature of 400 K and the initial temperature of the disc is 300 K. If the thermal conductivity of the material of the cylinder is 10 W/mK and the specific heat capacity of the material of the disc is 600 J/kg–K, how long will it take for the temperature of the disc to increase to 350 K? Assume, for purposes of calculation, the heat capacity of the cylinder to be very low and the system to be thermally insulated except for the upper face of the cylinder.

4. A hot body placed in air is cooled down according to Newton's law of cooling, the rate of decrease of temperature being k times the temperature difference between the body and the surrounding. Starting from t = 0, find the time in which the body will lose half the maximum heat it can lose.

5. A cylindrical rod of heat capacity 120 J/K at room temperature $27°C$ is heated internally by a heater of power 240 W. The steady state temperature attained by the rod is $37°C$. Now the heater is switched off. Assuming Newton's law of cooling to hold, find
(a) the initial rate of increase in temperature
(b) the steady state rate of emission of radiant heat.
(c) the rate of decrease in temperature when heater is switched off
(d) the rate of decrease in temperature of the cylinder when its temperature falls to $31°C$
(e) the total amount of heat lost by the cylinder

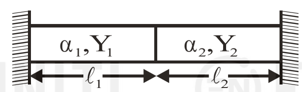
6. One end of a rod of length L and cross-sectional area A is in a furnace at temperature T_1. In steady state, the other end of the rod is kept at a temperature T_2. The thermal conductivity of the material of the rod is K and emissivity of the rod is e. It is given that $T_2 = T_0 + \Delta T$, where $\Delta T << T_0$ and T_0 is the temperature of the surroundings. If ΔT is proportional to $T_1 - T_2$, find the proportional constant. The rod is insulated and heat is lost only by radiation at the end where the temperature of the rod is T_2.



PARAGRAPH TYPE

Passage for Question Nons. 7 to 8

Two rods of equal cross sectional area are joined end to end as shown in figure. These are supported between two rigid vertical walls. Initially the rods are unstrained.



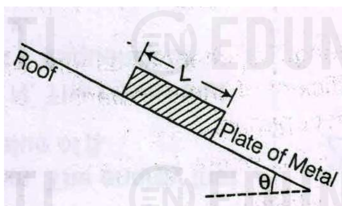
7. If temperature of system is increased by ΔT, then junction will not shift if
(a) $Y_1\alpha_1 = Y_2\alpha_2$ (b) $Y_1\alpha_1\ell_1 = Y_2\alpha_2\ell_2$
(c) $\alpha_1 = \alpha_2$ (d) $Y_2\alpha_1\ell_1 = Y_1\alpha_2\ell_2$

8. If temperature of system is increased by ΔT, then shifting in junction is given by (Assume that shifting is towards right)
(a) $\frac{\ell_1\ell_2(Y_1\alpha_2 - Y_2\alpha_1)\Delta T}{Y_1\ell_1 + Y_2\ell_2}$ (b) $\frac{\ell_1\ell_2(Y_1\alpha_1 - Y_2\alpha_2)\Delta T}{Y_1\ell_2 + Y_2\ell_1}$
(c) $\frac{\ell_1\ell_2(Y_1\alpha_1 - Y_2\alpha_2)\Delta T}{Y_1\ell_1 + Y_2\ell_2}$ (d) $\frac{\ell_1\ell_2(Y_1\alpha_2 - Y_2\alpha_1)\Delta T}{Y_1\ell_2 + Y_2\ell_1}$

Passage for Question Nons. 9 to 10

A perfectly plane house roof makes an angle θ with the horizontal surface. A flat square metal plate of side L rests over the roof. Each day, temperature varies between day's high (T_1) and day's low (T_2). Coefficients of thermal expansions of roof and plate are α_1 and α_2 $(\alpha_1 < \alpha_2)$ respectively. Coefficient of kinetic friction between plate and roof is μ_k.

Plate and roof are always in contact and both have same temperature at every instant of day. Because of difference in coefficients of thermal expansions, each bit of plate is moving relative to the roof, except for points along a certain horizontal line (called stationary line). The stationary line occupies no area, so we assume no force of static friction acting on the plate while the temperature is changing.



9. When the temperature is rising, then distance of the stationary line from top edge of plate will be

(a) $\frac{L}{2}\left(1-\frac{\tan\theta}{\mu_k}\right)$ (b) $\frac{L}{2}\left(1+\frac{\tan\theta}{\mu_k}\right)$

(c) $\frac{L}{2}\left(\frac{\tan\theta}{\mu_k}\right)$ (d) $\frac{L}{2}$

10. When the temperature is falling, then distance of the stationary line from top edge of plate will be

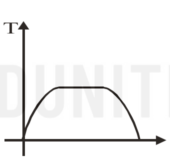
(a) $\frac{L}{2}\left(1-\frac{\tan\theta}{\mu_k}\right)$ (b) $\frac{L}{2}\left(1+\frac{\tan\theta}{\mu_k}\right)$

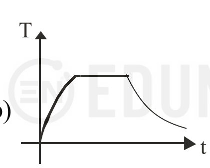
(c) $\frac{L}{2}\left(\frac{\tan\theta}{\mu_k}\right)$ (d) $\frac{L}{2}$

Passage for Question Nons. 11 to 13

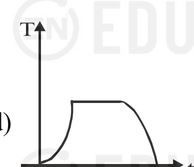
An electric kettle of unknown power rating is switched on after pouring 1 litre water in it. The temperature becomes constant at 60^0C after some time. The room temperature is 20^0C. When the kettle is switched off, during first 20 s, water cools down by 2^0C. Assume Newton's law of cooling to hold. (Specific heat of water is 4200 J/kg^{o}C)

11. Which is the best graph to represent temperature v/s time?

(a)


(b)


(c)


(d)


12. What is the wattage of the kettle

(a) 840W (b) 1W

(c) 100W (d) 420 W

13. What is the approximate time taken for the water to cool to 40^0C.

(a) 510 s (b) 270 s

(c) 120 s (d) 410 s

Passage for Question Nons. 14 to 16

In an experiment of study of Newton's law of cooling, a ball of metal is heated from inside by a heater of 20 W in a room maintained at constant temperature of $20°C$.

Following observations are made.

(i) Mass of the ball is 1 kg.

(ii) Rate of increase of temperature decreases with rise of temperature.

(iii) Temperature of ball becomes $50°C$ after some time and then remains constant.

(iv) Specific heat of metal is 500 $\mathrm{Jkg^{-1}\,°C^{-1}}$.

Corresponding to above observations, answer the following questions.

14. Assuming Newton's law of cooling holds for the ball, rate of heat radiated when the ball is at $30°C$ is

(a) (10/3) W (b) (20/3) W

(c) 10 W (d) (40/3) W

15. The time taken for the temperature to increase from $20°C$ to $30°C$ is (Take $\ln 1.5 = 0.4$)

(a) 400 s (b) 300 s

(c) 200 s (d) 600 s

16. During the period, the temperature of ball rises from $20°C$ to $30°C$, total heat energy radiated by the ball is

(a) 2000 J (b) 1500 J

(c) 750 J (d) 1000 J

Passage for Question Nons. 17 to 18

A body of area 0.8×10^{-2} m^2 and mass 50g directly faces the sun on a clear day. The body has an emissivity of 0.8 and specific heat of 3000 J/kg K. The surroundings are at $27°C$. (solar constant, $G_{SC} = 1.4\,\mathrm{kW/m^2}, \sigma = 5.67\times10^{-8}\ \mathrm{W/m^2K^4}$).

17. The rate of rise of the body's temperature (when its temperature is $27°C$) is nearly

(a) $0.36°C/s$ (b) $0.06°C/s$

(c) $0.18°C/s$ (d) $0.12°C/s$

18. The maximum attainable temperature of the body is

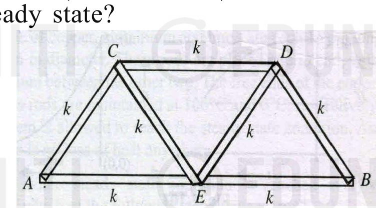
(a) $153°C$ (b) $396°C$

(c) $85°C$ (d) $40°C$

MORE THAN ONE CORRECT TYPE

19. Water contained in a jar at room temperature (20°C) is intended to be cooled by method-I or method-II given below
Method -I: By placing ice cubes and allowing it to float.
Method-II: By wrapping ice cubes in a wire mesh and allowing it to sink.
Choose best method(s) to cool the water.
(a) Method-I from 20°C to 4°C
(b) Method-I from 4°C to 0°C
(c) Method-II from 20°C to 4°C
(d) Method-II from 4°C to 0°C

20. Seven identical rods of material of thermal conductivity k are connected as shown in fig. All the rods are of identical length l and cross-sectional area A. If one end A is kept at $100\,°C$ and the other end B is kept at $0\,°C$, what would be the temperatures $(\theta_C, \theta_D \text{ and } \theta_E)$ of the junctions C, D and E in the steady state?



(a) $\theta_C > \theta_E > \theta_D$
(b) $\theta_E = 50°C$ and $\theta_D = 37.5°C$
(c) $\theta_E = 50°C, \theta_C = 62.5°C$ and $\theta_D = 37.5°C$
(d) $\theta_E = 50°C, \theta_C = 60°C$ and $\theta_D = 40°C$

21. A hollow copper sphere and a hollow copper cube of same surface area and negligible thickness, are filled with warm water of same temperature and placed in an enclosure of constant temperature, a few degrees below that of the bodies. Then in the beginning,
(a) the rate of energy lost by the sphere is greater than that by the cube
(b) the rate of energy lost by the two are equal
(c) the rate of energy lost by the sphere is less than that by the cube
(d) the rate of fall of temperature for sphere is less than that for the cube .

22. Three bodies A, B and C have equal surface area and thermal emissivities in the ratio $e_A : e_B : e_C = 1 : 1/2 : 1/4$. All the three bodies are radiating at the same rate. Their wavelengths corresponding to maximum intensity are λ_A, λ_B and λ_C respectively and their temperatures are T_A, T_B and T_C on kelvin scale. Then, select the correct statements.
(a) $\sqrt{T_A T_C} = T_B$
(b) $\sqrt{\lambda_A \lambda_C} = \lambda_B$
(c) $\sqrt{e_A T_A}\ \sqrt{e_C T_C} = e_B T_B$
(d) $\sqrt{e_A \lambda_A T_A . e_C \lambda_C T_C} = e_B \lambda_B T_B$